LTI transforms - repeated Eigenvalues

Suppose $\det(sI - A) = (s-\lambda_1)^r (s-\lambda_{r+1}) \cdots (s-\lambda_n)$

i.e. λ_1 repeated r times

$\lambda_{r+1}, \ldots, \lambda_n$ distinct

Then $\exists P$ s.t. $A_p = P^{-1} A P$ has form

$$A_p = \begin{bmatrix} [J_1(\lambda_1)] & & & \\ & [J_2(\lambda_1)] & & \\ & & \ddots & \\ & & & [J_\ell(\lambda_1)] \\ & & & & \lambda_{r+1} \\ & & & & & \ddots \\ & & & & & & \lambda_n \end{bmatrix}$$

"Jordan normal form"

where each J_i is an $n_i \times n_i$ "Jordan block"

$$J_i(\lambda) = \begin{bmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ & & & \ddots & 1 \\ 0 & & \cdots & & \lambda \end{bmatrix}$$

(n_i rows, n_i columns)

i.e. $A_p = \text{blkdiag}\{J_1, \ldots, J_\ell, \lambda_{r+1}, \ldots \lambda_n\}$

Constraint

$$\sum_{i=1}^{\ell} n_i = r$$

Terminology:

r = "algebraic multiplicity" of λ_1

ℓ = "geometric multiplicity"

Fact:

$\ell = \nu(\lambda_1 I - A)$ "nullity"

$=$ # of linearly indep eigenvectors associated with λ_1.

$= n - \rho(\lambda_1 I - A)$

= recall when all λ_k distinct can use $P = [v_1 \ldots v_n]$ to produce a purely diagonal A_p.

Special case:

(1) $\ell = \rho(\lambda_1 \mathbb{I} - A) = r$

$\Rightarrow$ r linearly indep eigenvectors for λ_1

Let $\underline{\omega}_1 \dots \underline{\omega}_r$ be these eigenvectors, and

$$P = [\underline{\omega}_1\ \underline{\omega}_2 \dots \underline{\omega}_r\ \underline{v}_{r+1} \dots \underline{v}_n]$$

Then A_p is purely diagonal

$$A_p = \begin{bmatrix} \lambda_1 & & & & \\ & \ddots & & & \\ & & \lambda_1 & & \\ & & & \lambda_{r+1} & \\ & & & & \ddots \\ & \emptyset & & & & \lambda_n \end{bmatrix}$$

(r times)

here $\ell = r$

and $\sum_{i=1}^{r} n_i = r \iff n_i = 1$

(2) $\ell = \rho(\lambda_1 \mathbb{I} - A) = 1$

$\Rightarrow$ only 1 Jordan block, must be $r \times r$

$$A_p = \begin{bmatrix} J_1(\lambda_1) & & & \\ & \lambda_{r+1} & & \emptyset \\ & & \ddots & \\ & \emptyset & & \lambda_n \end{bmatrix}$$

$$e^{A_p t} = \text{blkdiag}\{ e^{J_1(\lambda_1)t}, e^{\lambda_{r+1} t}, \dots, e^{\lambda_n t} \}$$

where:

$$e^{J_1(\lambda_1)t} = e^{\lambda_1 t} \begin{bmatrix} 1 & t & \frac{1}{2}t^2 & \cdots & \frac{1}{(r-1)!}t^{r-1} \\ 0 & 1 & t & \cdots & \frac{1}{(r-2)!}t^{r-2} \\ & & \ddots & \ddots & \vdots \\ & \emptyset & & \ddots & t \\ & & & & 1 \end{bmatrix}$$

(3) For $1 < \ell < r$, we know how many Jordan blocks, but not necessarily their sizes

$\Rightarrow$ Note for $\ell = 1$,

$$P = [\underline{v}_1 \; \underline{v}_{r+1} \cdots \underline{v}_n]$$

(2 + remaining)

columns come from a "chain of generalized eigenvectors", satisfying

$$(\lambda_1 \mathbb{I} - A)^K \underline{\omega} = \underline{0} \qquad K \geq 1, \; K \text{ integer}$$

= These "chains" generate, in a complicated fashion, the exact sizes of each block

we do at least have the constraint

$$\sum_{i=1}^{\ell} n_i = r$$

which can sometimes uniquely determine the structure of A_p:

i.e. $r = 4, \ell = 3$

$\Rightarrow n_1 + n_2 + n_3 = 4$

$\Rightarrow n_1 = 2, n_2 = n_3 = 1$

But if $r = 4, \ell = 2$

could have $n_1 = 3, n_2 = 1$

or $n_1 = n_2 = 2$

- which case we're in depends on exact details of A and is very sensitive to small changes in entries of A.

$$A = \begin{bmatrix} 1 & 1 \\ \varepsilon & 1 \end{bmatrix}$$

Calculation is very sensitive to even the smallest perturbation in A.

Repeated eigenvalue, $r=2$, $e=1$

if $\varepsilon = 0$ $\quad A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = A_p$

(non) repeated eigenvalues

if $\varepsilon \neq 0$ $\quad A_p = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$

$\lambda_1 \neq \lambda_2$

Matlab "Jordan" function useful; but doesn't always work.

= Time-varying transformations

recall

$$A_p(t) = P^{-1}(t)\left[A(t)P(t) - \dot{P}(t)\right]$$

Claim: for any $A(t)$, and any constant R, there exists $P(t)$ so that $A_p(t) = R$

i.e. (looking only at homog. dynamics)

$$\dot{x}(t) = A(t)x(t) \Longleftrightarrow \dot{z}(t) = Rz(t)$$

$P(t)$

=

Let $\Psi(t)$ be any fund'l matrix for $A(t)$

Let $P(t) = \Psi(t)e^{-Rt}$

Note $\det P(t) = \det\Psi(t) \det e^{-Rt}$

$\det\Psi(t) \neq 0$ for any finite t

$\det e^{-Rt} \neq 0$ for any finite t

$$\dot{P} = \left[\frac{d}{dt}\Psi(t)\right]e^{-Rt} + \Psi(t)\left[\frac{d}{dt}e^{-Rt}\right]$$

$\frac{d}{dt}\Psi(t) = A(t)\Psi(t)$

$\frac{d}{dt}e^{-Rt} = e^{-Rt}(-R)$

$$A_p = \left[\Psi(t)e^{-Rt}\right]^{-1}\left[A(t)\Psi(t)e^{-Rt} - A(t)\Psi(t)e^{-Rt} + \Psi(t)e^{-Rt}R\right]$$

$= R$ ✓

Example

$\dot{x} = x \Rightarrow A = 1$

Let $R = -1 = A_p \Rightarrow \dot{z} = -z$

$\Psi(t) = e^t \quad e^{-Rt} = e^t$

$P(t) = \Psi(t) e^{-Rt} = e^{2t}$

$\dot{P} = 2e^{2t}$

$A_p = P^{-1}[AP - \dot{P}]$

$= e^{-2t}[(1)e^{2t} - 2e^{2t}]$

$= -1 = R \checkmark$

$\Rightarrow \dot{z} = -z \Rightarrow z(t) = z_0 e^{-t}$

$x(t) = x_0 e^t$

$= P(t) z(t)$

$= e^{2t}[z_0 e^{-t}]$

$= z_0 e^t \checkmark$

Consistent, but seems "weird" that we can turn an unstable system into a stable one!

A "Lyapunov transformation" preserves properties like convergence or divergence.

$P(t)$ must also satisfy:

$\exists \eta > 0$ so that for all t

rules out "weird" behavior of previous example!

$|\det P(t)| \geq \eta > 0$

$|\det P^{-1}(t)| \geq \eta > 0$

=

"Floquet" Theory for LTP systems

"linear time periodic"

systems for which

$\exists T < \infty$ such that

$A(t+T) = A(t) \quad \forall t \geq t_0$

Will show: for such systems above "trick" can be applied while ensuring that $P(t)$ is a Lyapunov xform.